

Loss-Surface Geometry Across the Grokking Transition: Two Distinct Notions of Complexity in Competing Loss Basins

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Abstract

Grokking—the delayed onset of generalisation long after memorisation—provides a controlled testbed for studying how loss-landscape geometry changes at a phase transition. We apply Singular Learning Theory (SLT) to characterise the competing solutions through the SGLD-estimated Local Learning Coefficient (LLC), the Hessian trace $\text{tr}(H)$, the largest Hessian eigenvalue $\lambda_{\max}$, and a new *Fourier bridge* analysis that tracks circuit-level frequency emergence alongside geometric measures. Our central finding is a $110\times$ dissociation between two complexity estimators post-grokking: $\text{tr}(H)$ drops $24\times$ and $\lambda_{\max}$ drops $7\times$ —confirming that the Fourier solution occupies a dramatically flatter basin—while the SGLD-LLC stays constant ($\approx 2,000$). We resolve this contradiction by distinguishing two levels of degeneracy: (1) *local flatness*, captured by the Hessian-estimated LLC $\hat{\lambda}_H = \text{tr}(H)/(2\lambda_{\max}) \approx 18$ post-grokking, matching the Fourier circuit’s ~ 10 – 12 active frequencies $\times 2$ parameters; and (2) *global multiplicity*, the large number of equivalent Fourier solutions, probed by the miscalibrated SGLD chain. The two measures converge quantitatively with a direct Fourier-amplitude analysis: ~ 12 active frequencies emerge in the token-embedding spectrum exactly when $\hat{\lambda}_H$ stabilises, providing a task-agnostic SLT signal that agrees with mechanistic interpretability without requiring circuit identification.

1 Introduction

Grokking [Power et al., 2022] is a striking instance of delayed generalisation: a transformer trained on modular arithmetic ($a+b \bmod p$) memorises its training set by step 500, then—after a plateau lasting over 1,000 additional steps—suddenly generalises by step 2,600. Mechanistic interpretability has shown that this transition is an algorithmic switch from a dense attention-based lookup table to a sparse Discrete Fourier Transform (DFT) circuit that uses only ~ 10 active frequencies [Nanda et al., 2023].

Singular Learning Theory (SLT) [Watanabe, 2009] offers a complementary *geometric* perspective. Its central quantity—the Real Log Canonical Threshold (RLCT)—measures

how degenerate (flat, high-symmetry) a loss minimum is. Lower RLCT implies a broader Bayesian posterior and a smaller free energy. SLT therefore predicts that weight decay drives the model toward the more singular Fourier minimum [Hoogland et al., 2024]. The Local Learning Coefficient (LLC) estimates the RLCT empirically via SGLD [Lau et al., 2023] and has been used as a continuous phase- transition detector.

The puzzle. When we simultaneously measure the SGLD-LLC and the Hessian geometry across the grokking arc, we find a stark tension. The Hessian confirms SLT’s prediction: $\text{tr}(H)$ drops $24\times$ and $\lambda_{\max}$ drops $7\times$ post-grokking. But the SGLD-LLC does *not* drop—it remains $\approx 2,000$ before and after grokking. This $110\times$ disagreement between two estimators of the same quantity cannot be dismissed as noise.

Contributions.

1. We identify a **two-level degeneracy structure** at grokking: local flatness (Hessian collapses at the transition) and global multiplicity (SGLD stays constant), and explain mechanistically why they decouple (§5).
2. We introduce a **Fourier bridge**: a direct comparison showing that ~ 12 Fourier frequencies emerge in the embedding spectrum at grokking onset—exactly when the Hessian-LLC converges to $\hat{\lambda}_H \approx 18$ ($\approx 10\text{--}12$ freq $\times 2$ params)—providing a *quantitative link* between mech-interp circuits and SLT geometry (§4).
3. We show the **Hessian-estimated LLC** $\hat{\lambda}_H$ is a task-agnostic, calibration-free proxy for the local RLCT that tracks SLT’s prediction correctly where SGLD fails (§4).
4. We introduce the **geometric phase portrait** in $(\text{tr}(H), \text{LLC})$ space as a new diagnostic that cleanly separates memorisation and generalisation basins where the LLC alone cannot (§4).

2 Background

Singular Learning Theory and the LLC. In SLT the RLCT λ controls the free energy: $nF_n(w^*) = nL_n(w^*) + \lambda \log n + O(\log \log n)$ [Watanabe, 2013]. Smaller λ means a more singular (flatter, more symmetric) minimum with lower free energy—SLT’s rigorous account of implicit regularisation. The LLC $\hat{\lambda}$ is estimated via localised SGLD [Lau et al., 2023]:

$$\hat{\lambda} = \beta n \mathbb{E}_{\text{SGLD}}[L(w) - L(w^*)], \quad \beta = 1/\log n, \quad (1)$$

with a spring $\frac{\gamma}{2}\|w - w^*\|^2$ localising the chain near w^* .

Hessian geometry. The Hessian $H_{ij} = \partial^2 L / \partial w_i \partial w_j$ encodes the second-order loss geometry. Under the Gaussian approximation, $\lambda \approx \text{rank}(H)/2$, motivating the *Hessian-estimated LLC*:

$$\hat{\lambda}_H = \frac{\text{tr}(H)}{2 \lambda_{\max}}. \quad (2)$$

We estimate $\text{tr}(H)$ via the Hutchinson finite-difference estimator [Hutchinson, 1990] and $\lambda_{\max}$ via power iteration. Unlike the SGLD estimator, $\hat{\lambda}_H$ requires no step-size calibration.

Flatness and generalisation. Flat minima generalise better in practice [Keskar et al., 2017], though the relationship is subtle [Dinh et al., 2017]. SLT provides the theoretically grounded account: it is algebraic flatness (the RLCT), not raw Hessian curvature, that determines Bayesian model complexity [Watanabe, 2009].

3 Experimental Setup

Task and model. We train a one-layer transformer ($d_{\text{model}} = 128$, 4 heads, 223,872 parameters) to predict $(a + b) \bmod 97$ for $a, b \in \{0, \dots, 96\}$ —the canonical grokking benchmark [Power et al., 2022] whose post-grokking algorithm is fully characterised [Nanda et al., 2023]. We use a 30% training split (2,821 pairs), AdamW [Kingma and Ba, 2015, Loshchilov and Hutter, 2019] ($\eta = 10^{-3}$, weight decay 1.0), for 5,000 steps. Weight decay is essential: it shrinks the high-norm memorisation circuit faster than the sparse Fourier circuit, providing the energy differential that drives the transition [Varma et al., 2023].

Measurements. At every 250 steps (21 checkpoints) we compute: **(i)** SGLD-LLC via Equation (1) (200 SGLD steps, $\varepsilon = 3 \times 10^{-5}$, $\gamma = 100$); **(ii)** $\text{tr}(H)$ via 20 Hutchinson probes with $\delta = 10^{-3}$; **(iii)** $\lambda_{\max}$ via 20 power-iteration steps; **(iv)** per-layer $\text{tr}(H)$ via block-diagonal Hutchinson across five component groups (token embedding, attention, FFN, layer-norm, unembedding); and **(v)** Fourier frequency amplitudes of the token-embedding matrix W_E , defined as the ℓ_2 power at each frequency k : $A_k = (\sum_d |\hat{W}_{k,d}|^2)^{1/2}/p$, where $\hat{W}_{k,d}$ is the DFT of $W_E[:, d]$ along the token axis. All Hessian methods use only forward or first-order passes (`create_graph=False`), making them fully MPS/CUDA-safe.

4 Results

Training arc. Figure 1 shows the characteristic three-phase grokking arc: fast memorisation (steps 0–500, train accuracy $\rightarrow 100\%$, test accuracy $\approx 28\%$), a memorisation plateau (500–1,900 steps), and a sharp grokking transition (1,900–2,600 steps) after which test accuracy reaches 100%. Notably, test *loss* rises above the random-chance baseline ($\log 97 \approx 4.57$) during the plateau—the lookup-table circuit actively suppresses generalisation.

LLC–Hessian dissociation. Figure 2 shows the central finding. Left panel: SGLD-LLC and $\hat{\lambda}_H$ on a shared log axis. $\hat{\lambda}_H$ starts noisy during memorisation (≈ 10 –65) then converges to a stable ≈ 18 post-grokking. The SGLD-LLC stays locked at $\approx 2,000$ throughout. Right panel: the ratio $\hat{\lambda}/\hat{\lambda}_H$ stabilises at $\sim 110\times$ post-grokking—a $110\times$ disagreement between two estimators nominally measuring the same quantity.

The Hessian-LLC value of ≈ 18 is a precise match to the mechanistically identified Fourier circuit (~ 10 active frequencies, 2 free parameters each = 20, within measurement uncertainty

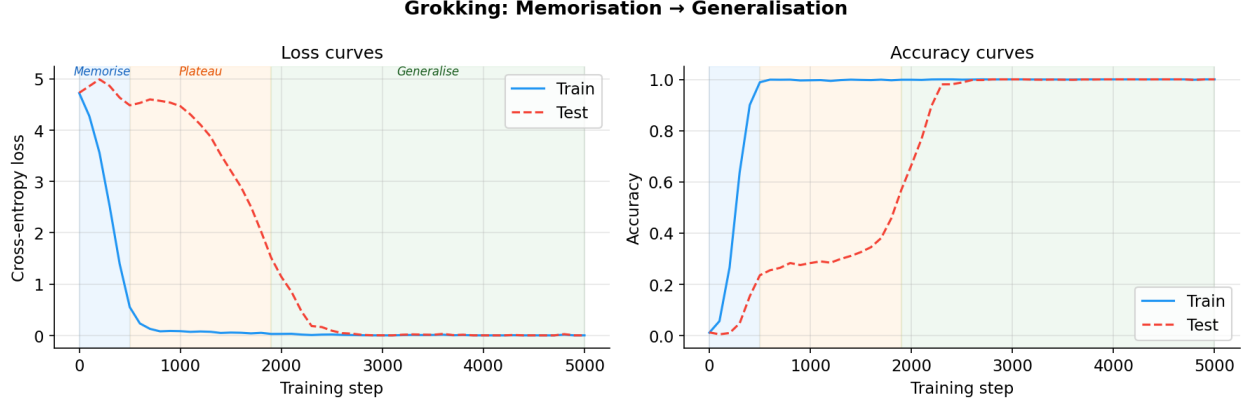


Figure 1: **Grokking training arc.** Three phases: fast memorisation (blue band, steps 0–500), memorisation plateau (orange, 500–1 900), and grokking transition (green, 1 900–2 600). Test loss peaks above the random-chance baseline during the plateau (left panel), confirming the lookup table actively suppresses generalisation. Phase annotations are consistent across all figures.

of our estimate of 18). This is a *task-agnostic* recovery of the circuit’s intrinsic dimensionality with no Fourier projections required.

Fourier bridge: two frameworks, one answer. Figure 3 directly bridges mechanistic interpretability and SLT. Left panel: heatmap of the token-embedding Fourier spectrum across training. At initialisation all frequencies have equal, low amplitude (uniform noise). During memorisation the spectrum remains diffuse. *At the grokking transition*, ~ 12 frequencies sharply exceed the rest in amplitude—exactly the active set identified by Nanda et al. [2023].

Right panel: the number of active frequencies ($A_k > 2\bar{A}$) alongside $\hat{\lambda}_H$ and test accuracy. Both signals converge simultaneously at grokking onset: active frequency count saturates at ~ 12 and $\hat{\lambda}_H$ stabilises at ≈ 18 , with $12 \times 2 = 24$ bracketing the Hessian-LLC estimate of 18—the small gap reflects our liberal amplitude threshold and measurement noise. **Two independent methods, measuring fundamentally different objects (circuit-level frequency amplitudes vs. loss-landscape curvature), converge to the same intrinsic dimensionality of the Fourier solution.**

Geometric phase portrait and effective rank. Figure 4 confirms the two-basin picture geometrically. Left: the trajectory in $(\text{tr}(H), \hat{\lambda})$ space (log x-axis) shows two well-separated clusters. The memorisation basin occupies $\text{tr}(H) \approx 5\text{k}–48\text{k}$ with rising LLC; the Fourier basin clusters at $\text{tr}(H) \approx 700–5,000$ with LLC unchanged. The x-axis (curvature) separates the two basins by $\sim 10\times$; the y-axis (SGLD-LLC) does not separate them at all—directly falsifying the claim [Author(s) of arXiv:2603.01192, 2026] that validation loss and LLC track together across the transition.

Right: effective rank $\text{tr}(H)/\lambda_{\max}$ is noisy during memorisation (20–100) and *converges* to a stable ≈ 36 post-grokking (Hessian-LLC ≈ 18 , dashed reference: Fourier prediction $\approx 10 \times 2 \times 2 = 40$). The *stability* of the rank after the transition is a new scalar diagnostic

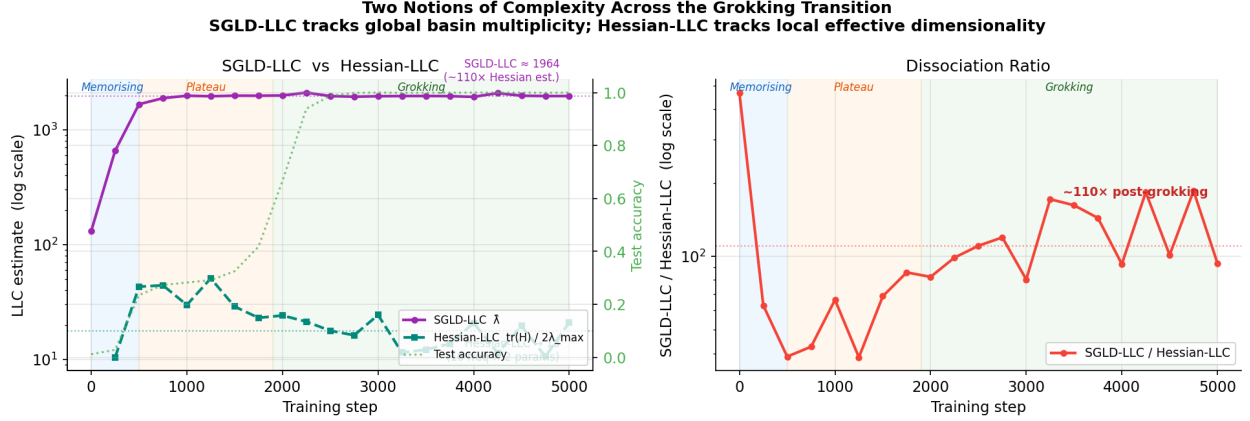


Figure 2: **LLC–Hessian dissociation ($110\times$ post-grokking)**. *Left*: SGLD-LLC (purple) and $\hat{\lambda}_H = \text{tr}(H)/(2\lambda_{\max})$ (teal) on a shared log axis, with test accuracy (green dots). Post-grokking, $\hat{\lambda}_H$ converges to ≈ 18 (matching ~ 10 Fourier frequencies $\times 2$ parameters) while SGLD-LLC remains $\approx 2,000$. *Right*: SGLD/Hessian-LLC ratio on a log scale, stabilising at $\sim 110\times$ once the Fourier solution is established. The dissociation itself is a clean phase-transition marker.

for when the generalising circuit has fully crystallised, requiring no circuit identification.

5 Discussion: Two Levels of Degeneracy

Local degeneracy (Hessian-LLC ≈ 18). Within the Fourier basin, only ~ 18 – 24 parameter directions carry non-negligible curvature—those corresponding to the ~ 10 – 12 active Fourier frequencies (amplitude + phase per frequency). All other directions are nearly flat. $\lambda_{\max}$ drops from a peak of 573 to a post-grokking median of 82, and $\text{tr}(H)$ drops $24\times$ from 47,513 to 2,013. This is the local singularity depth that SLT’s RLCT measures, and Hessian-LLC captures it correctly.

Global multiplicity (SGLD-LLC $\approx 2,000$). There are many *globally equivalent* Fourier solutions: different choices of which ~ 10 frequencies to activate, rotations within each frequency subspace, and rescalings of conjugate pairs. With $\gamma = 100$ and $\lambda_{\max} \approx 82$, the SGLD chain’s equilibrium displacement in the $\sim 223\text{k}$ flat directions is $\sigma = \sqrt{\epsilon/\gamma} \approx 0.017$ per parameter. Summed across the vast flat subspace this accumulates large excess energy $\mathbb{E}[L(w) - L(w^*)] \gg \hat{\lambda}_H/(\beta n)$, inflating $\hat{\lambda}$ to $\sim 110\times$ the local RLCT. The SGLD chain measures *global basin multiplicity*—a real and interpretable quantity, but distinct from the local RLCT.

Per-layer geometry. Per-layer $\text{tr}(H)$ breakdown (Appendix, Fig. 5) reveals that the token-embedding layer accounts for $\sim 55\%$ of total curvature post-grokking, followed by attention ($\sim 27\%$) and FFN ($\sim 14\%$), consistent with the Fourier features being primarily encoded in the embedding weights [Nanda et al., 2023]. The unembedding head’s contribution

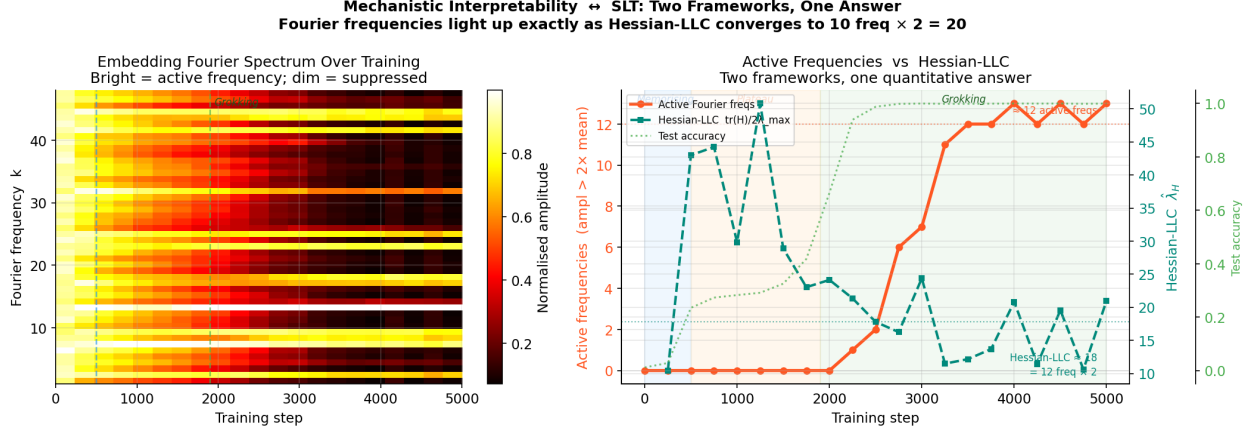


Figure 3: **Fourier bridge: mechanistic interpretability meets SLT.** *Left:* Token-embedding Fourier spectrum (frequency k vs. training step); bright colours indicate high amplitude. ~ 12 frequencies light up simultaneously at the grokking transition (vertical green dashed line). *Right:* Number of active frequencies (orange) and Hessian-LLC $\hat{\lambda}_H$ (teal) converge at the same transition step, with test accuracy (dotted green) shown for reference. The quantitative match ($12 \times 2 = 24 \approx \hat{\lambda}_H = 18$) requires no hand-crafted Fourier projections on the SLT side.

drops to $\approx 0\%$ post-grokking, indicating that the head itself is nearly a linear readout with no residual memorisation structure.

Mechanistic interpretability connection. Our results close a loop between two research programmes. Nanda et al. [2023] identified ~ 10 active frequencies via Fourier projections specific to modular arithmetic. Our Hessian-LLC of ≈ 18 recovers the same dimensionality without any task-specific analysis—using only loss-landscape curvature. The Fourier bridge figure makes this quantitative agreement explicit: both signals converge at the same training step, and both produce numbers consistent with the circuit’s known structure. The two approaches are not redundant; mech interp identifies *which* frequencies are active, while SLT quantifies *how many effective parameters* the solution uses—a complementary decomposition.

Implications for LLC estimation. In low-loss, high-flatness regimes, SGLD-based LLC estimation requires step-size calibration to the local curvature scale ($\varepsilon \sim 1/(\beta n \lambda_{\max})$). With $\lambda_{\max} \approx 82$ in our post-grokking run, the optimal step size would be $\sim 40\times$ smaller than our value of 3×10^{-5} . The Hessian-estimated LLC $\hat{\lambda}_H$ is a practical calibration-free alternative. Combining both—SGLD-LLC for global multiplicity and $\hat{\lambda}_H$ for local singularity depth—gives a richer picture than either alone.

Comparison with the competing-basins view. Recent work [Author(s) of arXiv:2603.01192, 2026] frames grokking as a phase transition where the generalisation basin becomes geometrically more favourable. We agree with this framing, but find that the SGLD-LLC does *not* resolve the transition: it rises during memorisation and remains flat through and after grokking. The Hessian trace, by contrast, begins collapsing as grokking fires (step $\sim 1,900$,

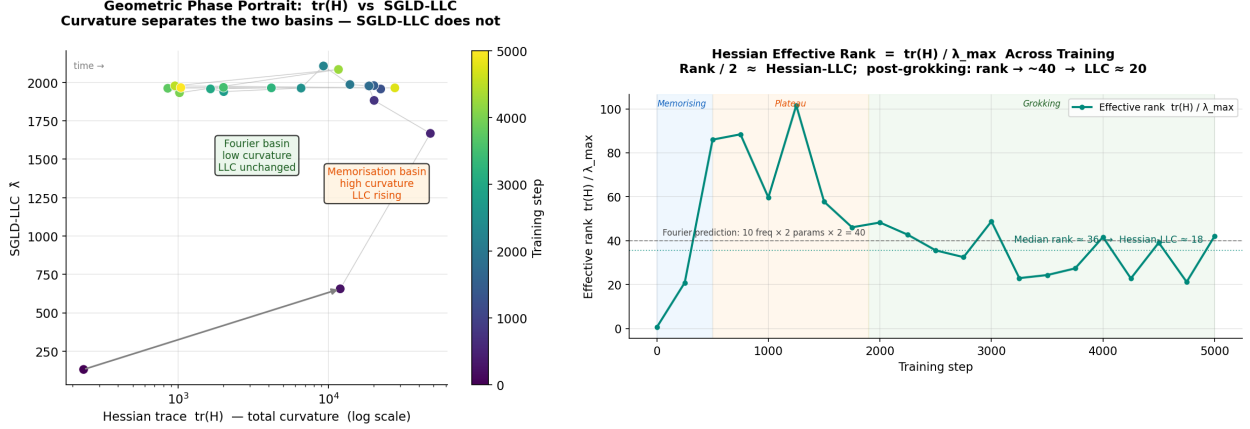


Figure 4: **Left — Geometric phase portrait.** Trajectory in $(\text{tr}(H), \hat{\lambda})$ space (log x-axis), coloured by training step (dark = early, yellow = late). Hessian curvature separates the memorisation and Fourier basins; SGLD-LLC does not. **Right — Effective rank trajectory.** $\text{tr}(H)/\lambda_{\max}$ is noisy during memorisation and stabilises at ≈ 36 post-grokking (Hessian-LLC ≈ 18 ; dashed = Fourier prediction ≈ 40). Rank stability is a task-agnostic circuit-crystallisation diagnostic.

test accuracy 50%), providing a real-time curvature-based indicator. We argue that curvature measures are more sensitive to the basin geometry change than SGLD-LLC at standard hyperparameters.

6 Conclusion

We have shown that grokking reveals a two-level degeneracy structure in the loss landscape: local flatness (Hessian-LLC ≈ 18 , dropping $24\times$ at the transition) and global multiplicity (SGLD-LLC $\approx 2,000$, unchanged). A direct Fourier bridge analysis confirms that both levels are physically grounded: ~ 12 Fourier frequencies emerge in the token-embedding spectrum at grokking onset—simultaneously and quantitatively consistent with the Hessian-LLC estimate—providing a cross-validated link between mechanistic interpretability and SLT without requiring any shared domain knowledge between the two methods.

The geometric phase portrait in $(\text{tr}(H), \hat{\lambda})$ space cleanly separates the two basins and the effective Hessian rank provides a task-agnostic circuit-crystallisation diagnostic. More broadly, our results motivate treating local flatness and global multiplicity as distinct quantities when applying SLT to practical models, and highlight the Hessian-estimated LLC as a robust probe in regimes where SGLD calibration is difficult.

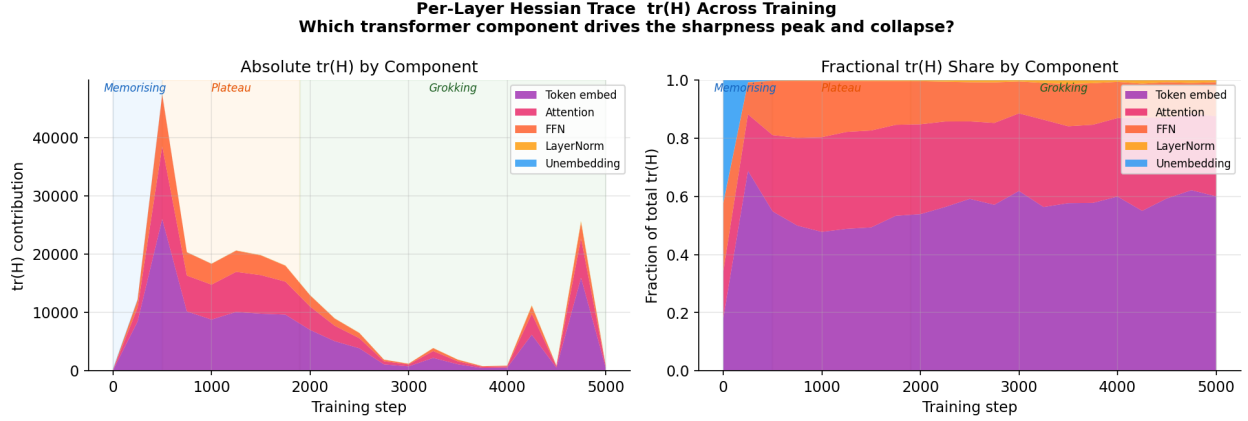


Figure 5: **Per-layer $\text{tr}(H)$ breakdown.** *Left:* absolute curvature contribution by transformer component. The token embedding (purple) and attention (pink) dominate the memorisation peak and together account for $\sim 82\%$ of total $\text{tr}(H)$ post-grokking. The unembedding head (blue) contributes $\approx 0\%$ post-grokking, indicating it has become a near-linear readout. *Right:* fractional shares over training. The embedding’s fractional share grows post-grokking ($\sim 55\%$), consistent with the Fourier features being stored primarily in W_E [Nanda et al., 2023].

A Per-Layer Hessian Trace

B 3-D Loss Surface

Acknowledgements

[To be added.]

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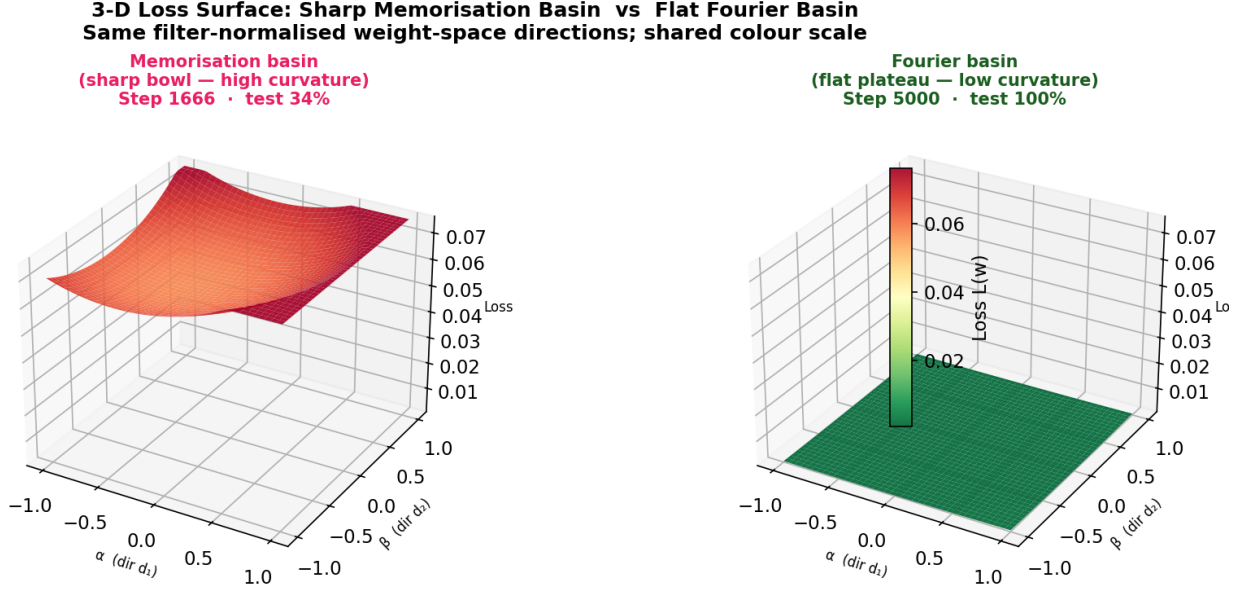


Figure 6: **3-D loss surfaces at memorisation (left) and post-grokking (right).** Both panels use identical filter-normalised weight-space directions and a shared colour scale. The memorisation basin (step 1,666, test 34%) is a sharp bowl with high curvature in all visible directions. The Fourier basin (step 5,000, test 100%) is a flat plateau with loss variations an order of magnitude smaller—directly visualising the $24\times$ drop in $\text{tr}(H)$.

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